The University of Texas at Austin

ADAPTIVE TOPOLOGY FOR HIERARCHICAL FEDERATED MOBILE LEARNING

Dec 5, 2024

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INTRODUCTION

Federated Learning (FL) traditionally operates in a centralized star topology, where individual devices train local models and send them to a central server for aggregation, leading to high communication overhead. Fully decentralized FL is inefficient because individual devices lack datasets representative of the global distribution, making aggregation challenging. Hierarchical (semi-centralized) FL schemes balance these extremes but often assume static network topologies, ignoring device mobility.

PROBLEM STATEMENT

Our project leverages device mobility to create an adaptive hierarchical FL scheme. We combine a top-down approach—adaptively assigning an efficient network topology inspired by weighted preferential attachment—with a bottom-up approach where mobile devices communicate with neighbors to form communities based on model similarity. This dual strategy aims to improve the efficiency of federated learning on a dynamic mobile network.

Unlike previous work that relies on fixed Wi-Fi access points or considers fully decentralized FL under static conditions, we dynamically select aggregation points among mobile devices and incorporate real-world network conditions. By allowing mobility and network conditions to inform community assignments, we enhance model aggregation within communities before global aggregation, adapting techniques from evolving network studies to suit our mobile FL context.

METHODOLOGY

We simulate mobile federated learning with 100 moving devices in Austin, each equipped with a convolutional neural network (CNN) and a unique partition of the CIFAR-10 dataset, reflecting a non-iid data distribution.

Hierarchical Aggregation Scheme (Figure 4)

- **Global Level:** The cloud server trains the global model and assigns a network topology optimized based on device locations.
- **Local Level**: Devices perform local aggregation at designated "access points" (APs).

Devices share model weights with neighbors, incurring communication costs proportional to **distance**. The server knows each device's location and AP models. We also further implemented various aggregation strategies such as FedAvg, FedProx, and FedMax. For this scenario results were obtained using only FedAvg.

Simulating Device Mobility

We use real-world mobility data from the **FourSquare** dataset over 10 days during peak hours (8 am–7 pm). Each coordinate from update aligns with a single global server round..

Algorithms for Network Optimization

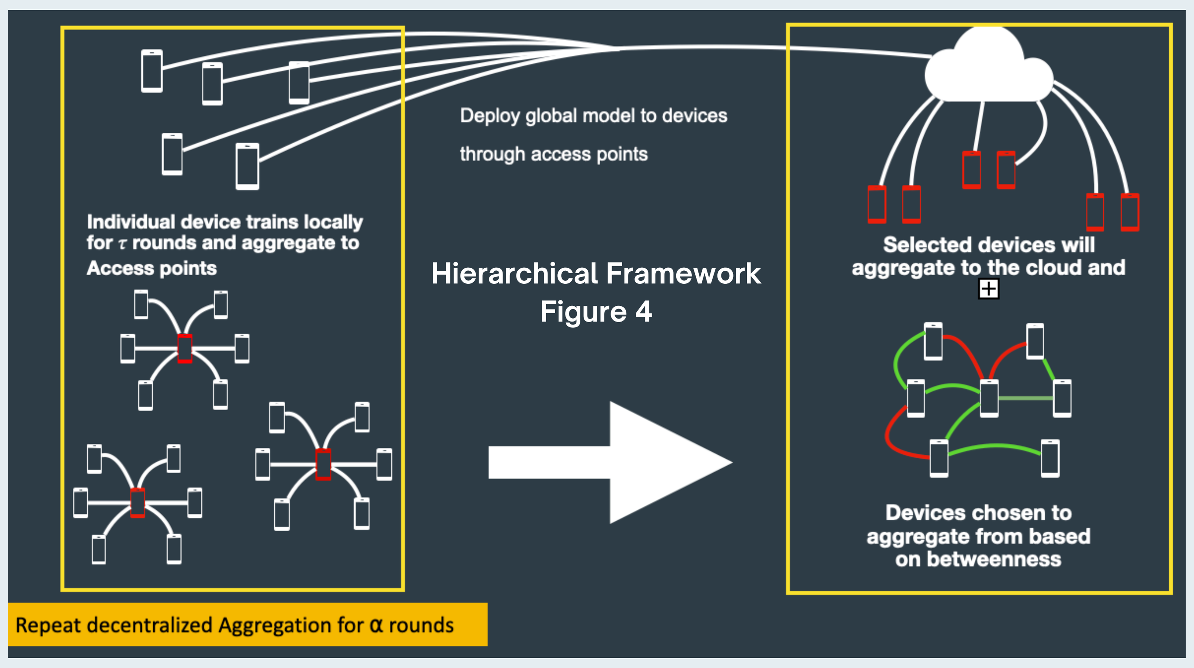
1. **Proximal Preferential Attachment (PPA)**
   - Purpose: Builds a scale-free topology for efficient communication.
   - Mechanism: During the linking phase, the server runs PPA, allowing each device to connect to a limited number of nearby devices.
2. **Cosine Reassignment**
   - Purpose: Groups devices with similar data distributions to help the non-IID challenge
   - Mechanism: Devices calculate cosine similarity between their model and community models, reassigning themselves to communities where similarity is highest.
3. **Dynamic Proximal Preference (DPP)**
   - Purpose: Combines PPA and Cosine Reassignment to cluster similar devices.
   - Mechanism: Constructs the graph like PPA but adjusts attachment weights using cosine similarity, promoting connections among devices with similar data. This can be seen in Algorithm 4.

Algorithm 4 Weighted-Preference Attachment

$D \leftarrow$ list of devices
for each device d in D **do**
  $\Pi \leftarrow$ empty list of weights
  for each proximity neighbor n of d **do**
    $\Pi \leftarrow \pi_n = \frac{\text{Degree}(n)+5\cdot\cos_{\text{sim}}(d,n)}{\text{Dist}(d,n)}$
  end for
  Attach k edges between d and neighbors sampled from Π
end for
return

OBJECTIVE

Formulate a new framework that is on par with or exceeds the performance of the traditional centeralized star FL with mobile devices



ANALYSIS & DISCUSSION

Training Accuracies

- PPA highlights mimicking scale-free improves performance (M1 insight)
- Cosine Sim struggles on its own (jagged learning curve) however ->
- Combining Cosine Sim to PPA improves performance (DPP)
- Finally, using FedMax (MaxDPP) slightly improves over DPP
- All semi-centralized algorithms train to higher accuracy in less overall server rounds than the central FL baseline **(Figure 1)**

Network Cost:

- **Table 1** compares the cost required to reach **45.56%** accuracy for each algorithm using two metrics:
  - Wall Clock Time: Total runtime accumulated over global rounds
  - Network Cost: Sum of communications over edges in our network which are weighted by the distance between incident nodes
- 45.56% is used as a baseline comparison since this was the maximum accuracy achieved by star in 35 rounds

Overall Performance:

- Despite faster training, no algorithm could surpass 55% accuracy.
- We are interested in exploring further ideas such as:
  - community memory for stable learning groups
  - hyper parameter search for optimal # of AP's and local rounds
  - further reduced costs with sparser network edges

MAIN RESULTS

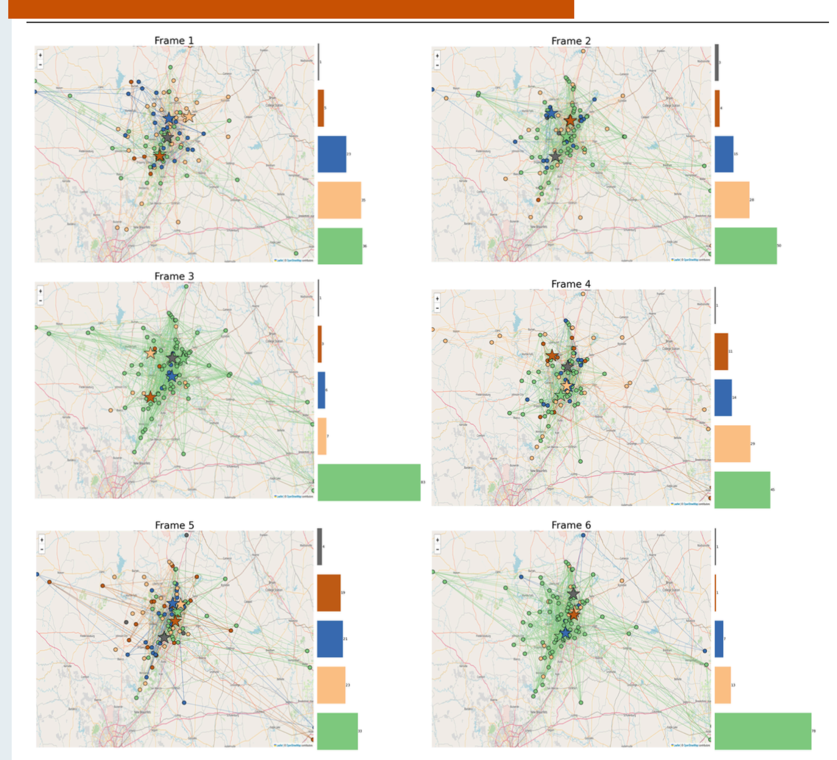


Figure 2: Device positions and network topology, color-coded by community assignment over the first 6 global training rounds for the DPP algorithm.

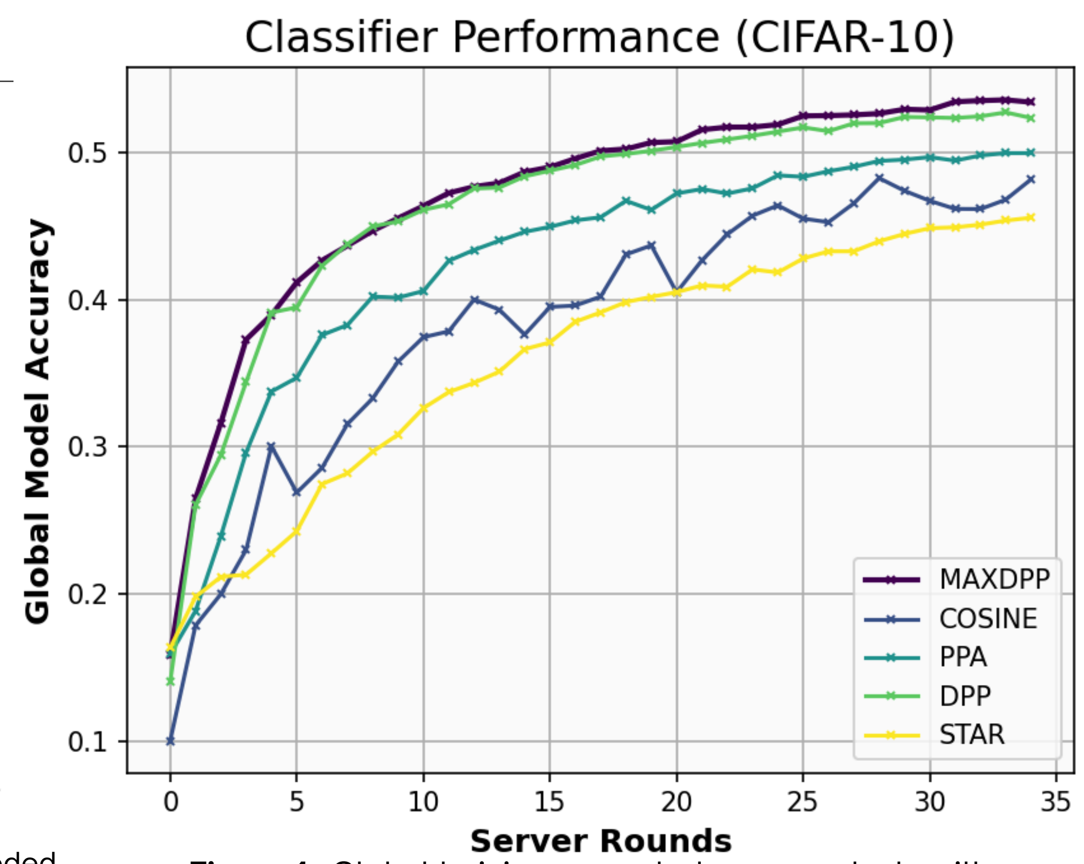
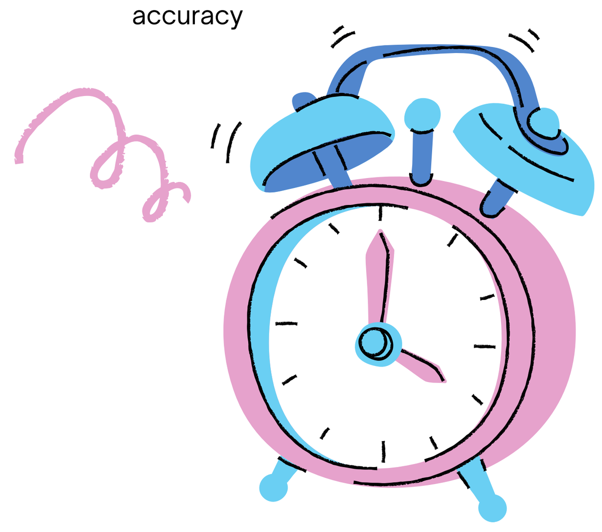


Figure 1 : Global training curve between each algorithm

	Wall Clock Time[s]	Network Cost
Star	691.22	3139.5
PPA	763.48	2938.02
Cosine	2497.14	146,744.5
DPP	402.75	1710.7
MaxDPP	**431.66**	**1602.064**

Table 1 : Cost and RunTime to reach 45.56% global accuracy



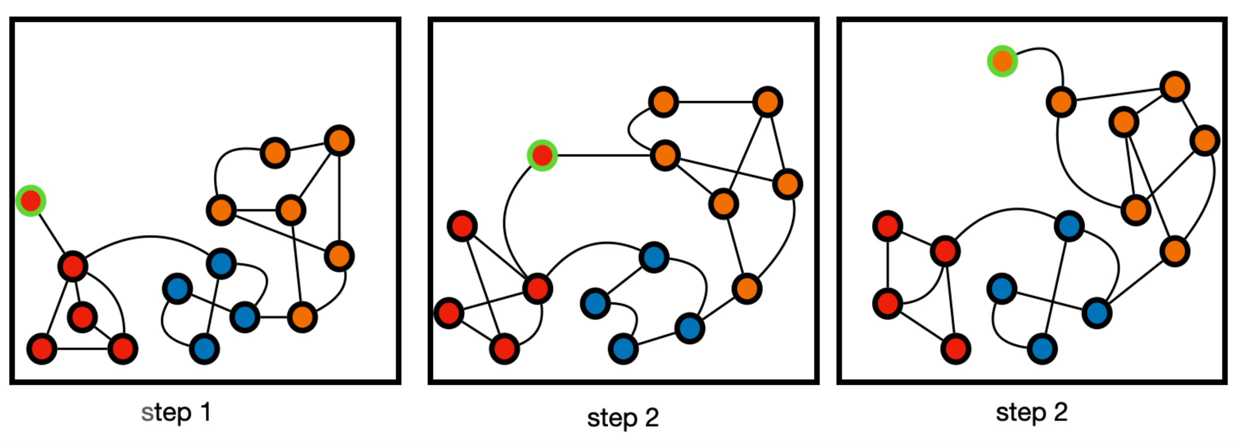


Figure 3: Weighted Proximal Preferential Attachment in mitigating an isolated node.

- Step 1, the node is assigned to the red community due to both proximity and model similarity.
- Step2, the node is in between the red and orange community, however due to model similarity from previous step, we would want it to be a part of the red community.
- Step 3, given the distance the node is from the red community, in order to save in network cost, we would like to have them be a part of the orange community.

CONCLUSION

Project Recap

- In Milestone 1 (M1), we found a scale free topology emulates central FL scheme in terms of training speed
- Inspired by this, we created an algorithm (PPA) which encourages hubs while considering proximity of mobile devices for M2
- Also in M2, we investigate a decentralized community assignment process to group devices based on model similarity
- Combining these 2 components, we get an algorithm (DPP), which trains faster in less global rounds than a centralized FL baseline, thus achieving our project's goal
- In M3, we increase model personalization by introducing FedMax aggregation improving performance slightly

Takeaways

- Our project uses network science to formulate a new hierarchical FL scheme capable of outperforming naive FL baselines and verify this on a real-world mobility dataset
- Future extensions would consider improving the community personalization for stable training groups, to further explore our idea of using model weight similarity and modularity algorithms

Related literature

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